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United States Patent	12414579
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Wenger; LaVon et al.

Food processing system including extruder with hollow core screw assembly

Abstract

Apparatus and methods for food production including a food preconditioner (228) operable to heat and partially pre-cook food ingredients, and a twin screw extruder (20) operable to further cook the preconditioned ingredients to create final food products. The extruder (20) includes a pair of hollow core extrusion screws (50, 52, 124, 126, 190) having elongated hollow core shafts (54, 128, 130, 192) equipped with helical fighting (56, 132, 134, 194) along the lengths thereof. The fighting (132, 134, 194) is also of hollow construction which communicates with the hollow core shafts (54, 128, 130, 192). The fighting (56, 132, 134, 194) also includes forward, reverse pitch sections (64, 162, 216). The extrusion screws (50, 52, 124, 126, 190) are designed to impart high levels of thermal energy into materials being processed in the extruders (20), without adding additional moisture.

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Appl. No.: 17/375757

Filed: July 14, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210337831 A1	Nov. 04, 2021

Related U.S. Application Data

Publication Classification

Int. Cl.: **A23K40/00** (20160101); **A23K10/20** (20160101); **A23K40/25** (20160101); **A23K50/40** (20160101); **A23N17/00** (20060101); **A23P30/20** (20160101); **B01F27/1143** (20220101); **B01F27/2121** (20220101); **B01F27/2122** (20220101); **B01F27/27** (20220101); **B01F27/272** (20220101); **B01F27/722** (20220101); **B01F27/723** (20220101); **B01F27/726** (20220101); **B01F35/95** (20220101); **B29C48/40** (20190101); **B29C48/84** (20190101); B01F35/90 (20220101); B01F101/18 (20220101)

U.S. Cl.:

CPC **A23K40/25** (20160501); **A23K10/20** (20160501); **A23K50/40** (20160501); **A23N17/002** (20130101); **A23N17/004** (20130101); **A23N17/007** (20130101); **A23P30/20** (20160801); **B01F27/1143** (20220101); **B01F27/2121** (20220101); **B01F27/2122** (20220101); **B01F27/2721** (20220101); **B01F27/2722** (20220101); **B01F27/276** (20220101); **B01F27/722** (20220101); **B01F27/723** (20220101); **B01F27/726** (20220101); **B01F35/95** (20220101); **B29C48/402** (20190201); **B29C48/845** (20190201); B01F2035/99 (20220101); B01F2101/18 (20220101)

Field of Classification Search

CPC: B01F (27/2721); A23N (17/002); A23N (17/004); A23N (17/007)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a division of U.S. application Ser. No. 16/446,194 filed Jun. 19, 2019, which is a continuation of U.S. application Ser. No. 15/699,642 filed Sep. 8, 2017 (now U.S. Pat. No. 10,434,483 issued Oct. 8, 2019), which claims the benefit of provisional application Ser. No. 62/459,215, filed Feb. 15, 2017, each of which is incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention is broadly concerned with high thermal transfer, twin screw processing devices (e.g., extruders), which can be used in the production of comestible products having high meat contents. More particularly, the invention is concerned with such devices, as well as the twin screws therein, which provide very high thermal energy inputs without the need for direct injection of steam into the materials being processed. This is obtained by unique, hollow core, high thermal transfer screw designs which are equipped with steam injection apparatus for indirect heating of materials during processing thereof.

Description of the Prior Art

(2) Many pet foods are produced using extrusion technology, where in mixtures containing grains, starches, fats, and other ingredients are initially preconditioned to heat and partially cook the mixtures, followed by processing through a single or twin screw extruder. Preconditioning involves passage of the initially dry mixture into an elongated housing where it is mixed with injected steam and/or water, making the mixture better conditioned for downstream extrusion. During extrusion, it is common to inject still further quantities of steam and/or water into the mixture during passage through the extruder barrel. For many product recipes, this existing technology is satisfactory. However, in recent years, producers have sought to incorporate greater and greater quantities of fresh meat into the feed recipes. At relatively low levels of meat addition, the traditional extrusion processing technologies are suitable. However, when attempts are made to incorporate high levels of meat (e.g., greater than about 40% by weight), standard preconditioner/extruder equipment may not be adequate. One issue is that fresh meat products are very high in moisture, and thus the usual steps of moisture addition into the preconditioner and/or extruder renders the product too wet for successful extrusion.

(3) The following references are of interest: U.S. Pat. Nos. 3,255,814, 3,259,374, 3,386,708, 3,637,069, 3,776,529, 4,040,768, 4,372,734, 5,074,057, 5,547,277, 6,099,159, and 9,321,190, and French Patent No. FR2477429.

SUMMARY OF THE INVENTION

(4) The present invention overcomes the problems outlined above and provides twin screw processing devices, preferably but not necessarily in the form of extruders, having restricted orifice die structures. Generally speaking, the processing devices of the invention include an elongated barrel having a material inlet and a processed material outlet, with a pair of elongated, intermeshed, axially rotatable, helical screws within and extending along the length of the barrel. Each of the screws has an elongated shaft with a hollow core and an elongated helical fighting section extending outwardly from the shaft, with the hollow core configured to receive heat exchange media. The devices also include a media delivery assembly having a pair of media delivery tubes extending from a point outside of the barrel and respectively into a corresponding one of the hollow cores; structure such as rotary unions is operably coupled with the delivery tubes so as to supply media to the tubes for delivery into the hollow cores.

(5) In preferred forms, each of the helical screws has a first helically flighted section operable to convey material from the material inlet toward and through the processed material outlet, and a

second helically flighted section proximal to the processed material outlet, which is operable to retard the flow of material therepast, with the first and second helically flighted sections being of opposite hand. In such designs, it is preferably that the flow-retarding second helical section has a pitch length less than the pitch length of the first helical section.

(6) In order to provide the greatest degree of heat transfer from the hollow core screws, the latter are designed with a central section, which may be of integral, cast construction, with both a hollow core and a hollow screw in communication with each other. In an embodiment, the helical screw of the central section has a pair of opposed, spaced apart wall segments extending outwardly from the shaft and presenting a helical transition area between the opposed wall segments. This transition area is open throughout the length and width thereof to afford unobstructed communication between the hollow core and hollow screw. Moreover, the thickness of the hollow shaft and the hollow screw are substantially equal.

(7) The screws of the invention typically have an elongated, axially rotatable shaft with a hollow core and presenting a rear connection end operable to couple the screw with a drive, and forward bearing structure at the opposite end of the shaft. An axial bore extends through the forward end of the shaft, permitting insertion of a heat exchange media delivery tube through the bore and into the hollow core. Furthermore, the screws have elongated, helical fighting extending outwardly from the shaft along the length thereof, the fighting including a first section and a second, shorter section. The first and second fighting sections are of opposite hand, with the second section located between the first section and the bearing structure, and have an axial length at least about three times greater than the axial length of the second section.

(8) In another embodiment, the screws are manufactured with an inwardly extending helical, open-top groove along the length of the fighting, thereby defining a bottom wall between the groove and the shaft core. A plurality of spaced apart apertures are provided through the bottom wall or cover piece in order to communicate the helical groove and the shaft core, and the groove open top is closed using a welded-in-place helical cover.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a twin screw processing device in accordance with the invention;

(2) FIG. 2 is a fragmentary view of the twin screw assembly forming a part of the device of FIG. 1;

(3) FIG. 3 is a fragmentary view with parts broken away of the forward outlet end of the device of FIG. 1, depicting the reverse flight sections of the twin screws;

(4) FIG. 4 is a partial vertical section taken along line 4-4 of FIG. 5, illustrating the internal construction of the processing device;

(5) FIG. 5 is a vertical sectional view taken along line 5-5 of FIG. 4;

(6) FIG. 6 is a fragmentary view with parts broken away of the forward end of another twin screw processing device in accordance with the invention, depicting the reverse flight sections of the twin screws;

(7) FIG. 7 is a fragmentary view of the twin screw assembly forming a part of the device of FIG. 6;

(8) FIG. 8 is a sectional view of the device of FIG. 6, illustrating the full configuration of the twin screw assembly thereof;

(9) FIG. 9 is a vertical sectional view taken along line 9-9 of FIG. 8;

(10) FIG. 10 is a partial sectional view illustrating the hollow shaft/hollow helix construction of the twin screws of the device of FIG. 1;

(11) FIG. 11 is a fragmentary, perspective, exploded view illustrating the drive connection and steam diffuser forming a part of the twin screw assembly of the FIG. 6 embodiment;

- (12) FIG. 12 is a perspective view of another hollow core screw embodiment in accordance with the invention;
- (13) FIG. 13 is a fragmentary elevational view of the screw of FIG. 12;
- (14) FIG. 14 is a fragmentary, vertical sectional view of the screw of FIG. 12, illustrating the internal construction thereof;
- (15) FIG. 15 is a fragmentary, enlarged cross-sectional view of the screw of FIG. 12;
- (16) FIG. 16 is another fragmentary, enlarged cross-sectional view of the screw of FIG. 12;
- (17) FIG. 17 is a vertical sectional view taken along the line 17-17 of FIG. 14;
- (18) FIG. 18 is a vertical sectional view taken along the line 18-18 of FIG. 14; and
- (19) FIG. 19 is a schematic block diagram illustrating a processing system in accordance with the invention for the production of high-meat content pet foods.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (20) Turning now to the drawings, and particularly FIGS. 1-5, a processing device 20 is illustrated. Broadly speaking, the device 20 includes an elongated, tubular barrel 22 having a material inlet 24 and an opposed processed material outlet 26, with a twin screw assembly 28 within the barrel 22
- (21) As illustrated, the barrel is made up of a plurality of end-to-end interconnected barrel sections 30, 32, 34, and 36. An internal liner assembly 38 is located within and extends along the length of the sections 30-36 and presents a pair of elongated, juxtaposed communicating arcuate passageways 40, 42, which receive the twin screw assembly 28. The sleeve further presents the rearmost opening 44, as best seen in FIG. 4. An apertured front wall 46 is located at the opposite end of the barrel and is secured thereto. Additionally, a forwardly projecting bearing housing 48 is attached to the outer face of front wall 46, and has a pair of side-by-side tubular bearings therein.
- (22) The twin screw assembly 28 includes identical, complementary first and second intermeshing, double-flight screws 50 and 52, which are designed to directionally co-rotate during the operation of device 20. Referring to FIGS. 2 and 4, it will be seen that the screws 50, 52 each have an elongated shaft 54 with outwardly extending, helical fighting 56 along the length of the shaft 54, having a pitch length of 1, based upon the screw diameter. An elongated central bore 58 extends substantially throughout the length of the shaft 54, thereby creating a hollow core 60 therein. As best seen in FIG. 3, the fighting 56 is divided into two sections, namely a first section 62 operable to convey material from the inlet 24 toward and through the processed material outlet 26, and a second section 64 operable to retard the flow of material therepast. To this end, the fighting sections 62, 64 are of reverse hand, respectively. The rearmost ends of the screws 50, 52 are provided with drive and bearing structure 66, 68, having rearmost, splined connector ends 70, 72; the ends 70, 72 are designed to mate with suitable drive structure (not shown) for axially rotating the screws. The forward ends of the screws are equipped with elongated extensions 74, 76, which are received within the tubular bearings of housing 48 (FIG. 3).
- (23) The screws 50, 52 are equipped with structure 78 to supply heat exchange media to the internal cores thereof. Specifically, elongated, stationary steam delivery tubes 80 and 82 extend from a point outside of housing 22 adjacent front wall 46 and into the bores 58. The tubes 80, 82 extend through the endmost tubular fixtures 84, 86 located at the forward ends of the extensions 74, 76. The outermost ends of the tubes 80, 82 connect with identical rotary unions 88, 90. Each such union includes a block 92 having a media inlet opening 94, a liquid removal outlet 96, and a rotatable sleeve 98 disposed about the corresponding tube 80 or 82. To this end, the sleeves 98 are secured to the corresponding tubular fixtures 84, 86.
- (24) The screws 50, 52 are preferably fabricated from metal using normal machining techniques, and are normally case-hardened.
- (25) In operation, material to be processed (which may be preconditioned, as explained hereafter) is delivered to inlet 24 during co-rotation of the screws 50, 52, which serves to advance the material during processing thereof to deliver processed material to outlet 26. During this operation, heat exchange media (normally steam) is directed from a delivery conduit (not shown) coupled with the

media inlets **94** and into the tubes **80, 82** to the open cores **60** of the screws in order to provide thermal energy required for processing of the material. Additional energy input is provided by means of pressure and sheer conditions developed within the barrel **22**. During processing, steam condensate passes from the cores **60** and through the sleeves **98**, exiting the system via the removal outlets **96**. In order to prevent leakage of material past the wall **46** and the fixtures **84, 86**, the reverse flight section **64** comes into play by creating a retardation force against the flow of material created by the opposite hand fighting of the section **62**. The product emerging from the outlet **26** may then be passed through a restricted orifice extrusion die and cut, in order to form the final product, although in practice, elongated delivery pipe is normally secured to the outlet **26** and a final die and knife assembly are located at the opposite end of the pipe.

(26) FIGS. **6-11** illustrate another embodiment of the invention, in the form of a processing device **100**. The device **100** is in many respects similar to the device **20**, having a tubular barrel **102** with a twin screw assembly **104** therein, and having an inlet **24** and an outlet **26**, as in the case of the first embodiment. Again, the barrel **102** is made up of tubular, interconnected barrel sections **106-112** with an internal liner assembly **114** defining side-by-side passageways **116, 118**. The forward end of barrel **102** is equipped with a front wall **120**, which supports a forwardly projecting bearing housing **122**.

(27) The screw assembly **104** has a pair of identical, intermeshed, single flight, helical screws **124, 126**, which are received within the passageways **116, 118**. Each of the screws has an elongated central shaft **128, 130**, as well as outwardly extending helical fighting **132, 134** along the length thereof. As in the case of the first embodiment, the screws **124, 126** have rearmost drive and bearing structures **136, 138**, equipped with splined drive connection ends **140, 142**. The forward ends of the screws have bearing extensions **144, 146**, which are received within the bearings of housing **122**.

(28) The principal differences between the first and second embodiments is the configuration of the helical screws **124** and **126**. Specifically, each such screw includes a primary central section **148** operable to move product along the length of the barrel **102** towards and through the outlet **26**, an inlet section **150** operable to deliver incoming material to the section **148**, and a forward, material flow retarding section **152**. The inlet section **150** is advantageously a machined part having a central shaft **154** with an internally splined bore **156**, and outwardly extending fighting **132, 134**. As best seen in FIG. **10**, the bore **156** is designed to receive the forward end of the corresponding drive and bearing structure **136, 138**. The screw sections **148** and **150** have a pitch length of **1**, based upon the screw diameter, and denoted by **D1** of FIG. **10**. The forward section **152** of each screw is likewise a machined part and has a central, bored shaft **160** with outwardly extending helical fighting **162**. Notably, the pitch of the flight **162** is opposite that of the pitch of the fighting **158** of screw section **148**, and has a pitch length of **0.3**, based upon the screw diameter, denoted by **D2** of FIG. **10**. Preferably, the pitch length **D1** should be from about **0.4-1.2**, more preferably **0.5-1.0**, whereas the pitch length **D2** should be from about **0.2-1.1**, more preferably **0.3-1**.

(29) The central section **148** is preferably formed by casting (e.g., sand or investment casting) with a plurality of sections which are butt-welded together to form the entirety of the central section. As best seen in FIG. **10**, the central section **148** has a central shaft **166**, which is hollow along the length thereof to define a central core **167**, as well as outwardly extending fighting **168**, which is likewise hollow to define a helical core **169**. In this regard, the flight **168** is defined by outwardly extending, opposed, spaced apart sidewalls **170** and **172**, with a flattened outer wall **174**. As such, it will be observed that there is a helical transition **176** between the innermost ends of the sidewalls **170, 172**, which provides full, open communication between the central core **167** and the helical core **169**, without any blockage or narrowing whatsoever; stated otherwise, the transition **176** presents an area which is open throughout the length and width thereof in order to afford unobstructed communication between the hollow regions of the shaft and helical screw. Inasmuch as the section **148** is of cast construction, it will be seen that the thickness of the shaft **166** is

essentially identical with the thicknesses of the sidewalls **170**, **172**, and outer wall **174**.

(30) In preferred forms, the length of the central screw section **148** is at least about three times, more preferably at least about five times, greater than the length of the forward screw section **152**.

(31) The screws **124**, **126** receive elongated media delivery tubes **178**, **180**, which are designed to deliver media such as steam to the interior of the screw sections **148** and **152**. The forward ends of the tubes **178**, **180** are received within rotary unions **88**, **90**, identical to those described in connection with the first embodiment, and like reference numerals have been applied (FIG. **10**).

The rotatable sleeves **98** are supported by couplers **84**, again as described in the first embodiment.

(32) The innermost ends of the tubes **178**, **180** are each supported by a tubular, open-sided diffusion cage **182**. The latter includes a mounting screw **184**, which extends through the end of the primary screw section **148** and is received within a threaded bore **186** in the forward butt end of bearing and drive structure **136**.

(33) The rearmost end of the screw section **148** has a splined portion **188**, which receives the forward end of the structure **136** ahead of inlet screw section **150**. The forward end of the screw section **148** is secured to the rearward end of screw section **152** by butt welding or any other appropriate technique. Hence, driving rotation of the structures **136** serves to rotate the entire screws **124**, **126**.

(34) The operation of device **100** is similar to the device **20**. However, owing to the completely open hollow core structure of the screw section **148**, better heat transfer from the injected steam is afforded, as compared with the device **20**.

(35) FIGS. **12-18** illustrate another hollow core screw embodiment of the invention in the form of helical extrusion screw **190**. The screw **190** is designed for use in a twin screw extruder, so that a mating screw (not shown) will be used in conjunction with the screw **190** to make a screw set. Generally, the screw **190** includes an elongated central shaft **192** with a continuous helical fighting **194** along the length thereof.

(36) The shaft **192** has a rearmost splined section **196** to afford a driving connection with a motor/gear reducer assembly, and a forward bearing extension. The shaft **192** is a machined, case-hardened part and has a solid rear section **198** and a hollow core forward section **200** presenting an elongated, axially extending, central core **202**. The forward end of the core **202** is equipped with a coupler **204** designed to receive a rotary union **88** (FIG. **14**). A stationary steam delivery tube **208** (shown fragmentarily in FIG. **14**) extends substantially the full length of the core **202** and has an open end **210**.

(37) The fighting **194** includes a rear section **212** of relatively narrow flight width, which extends the full length of the solid section **198**. Additionally, the fighting **194** has a wider flight width forward section **214** presenting an outermost flight surface **214a**, which extends from the end of section **212** to a point close to the forward end of the shaft **192**. However, as in the case of the earlier embodiments, the screw **190** has a reverse flight section **216** between the end of section **214** and coupler **204**.

(38) In the manufacture of the screw **190**, the fighting **194** is machined as a solid protrusion from the shaft **192**, with a continuous, helical, open-top groove **218** in the wide flight section **214**, extending from the outermost flight surface **214a** inwardly to an inner wall **220** close to the core **202**. Thereafter, a series of spaced apart apertures **222** are formed along the length of the inner wall **220**, in order to communicate the core **202** with groove **218**. Next, a helical cover piece **224** is positioned over the upper end of the groove **218**, and is welded to the fighting section **214**. In the final step, the screw **190** is machined to provide the proper outside diameter for the fighting **194**. This creates a unitary construction, as illustrated in the drawings.

(39) The operation of the screw **190**, with its mating, intermeshed screw within an extruder barrel, such as barrel **22**, is the same as described in connection with the embodiment of FIGS. **1-5**. That is, co-rotation of the screw set serves to advance material during processing thereof from the barrel inlet to the barrel outlet. Simultaneously, steam or other heat exchange media is directed into the

core **202** through the union **88** and the extension of shaft **192** beyond the end of the extruder barrel. This media flows through the core **202** and groove **218** owing to the communicating apertures **222**. This provides an increased level of thermal energy to the process. The reverse flight section **216** also serves to retard the flow of material at the forward end of the screw **190**.

(40) FIG. **19** schematically illustrates a system **226** for the production of high-meat content pet foods, and broadly includes an upstream preconditioner **228**, and an extruder **230**, the latter being a modified version of the devices **20** or **100**, using any of the hollow core screws described above. In particular, the extruder **230** includes a generally L-shaped conduit or pipe **232** affixed to outlet **26** and having an endmost, restricted orifice extrusion die **234**.

(41) The system **226** is designed to produce products using mixtures of meat with other conventional pet food ingredients such as grains, starches, and fats, together with minor ingredients such as vitamins and emulsifiers. The meat content of the starting mixtures typically ranges from about 100-250% by weight, more preferably from about 125-200% by weight, based upon the weight of the dry ingredients taken as 100% by weight. During processing, the product should meet a minimum temperature of 90° C. in order to meet feed safety requirements. The rotation of the twin screws in the range from about 30-150 rpm, more preferably from about 30-65 rpm; pressure within the housing is normally from about 200-600 psi, more preferably from about 300-400 psi. Pressurized steam is directed into the hollow core screws usually at a level of 30-60 psi, more preferably about 45 psi. Although not illustrated in the drawings, it is possible to employ extruder barrels which are jacketed for receiving steam or other heat exchange media.

(42) The preconditioner **228** may take a variety of forms, so long as the starting mixtures are heated and partially precooked prior to delivery to the inlet **24**; it is also possible to add moisture during preconditioning, but this is normally minimized or eliminated, owing to the high moisture content of the meat fraction of the mixtures. For example, it is possible to mix the dry ingredients in a conventional ribbon mixer or the like, followed by the addition of meat preheated to a temperature of from about 35-50° C. (more preferably about 45° C.) to the dry ingredients, with additional mixing. Alternately, use may be made of existing extrusion preconditioners, such as commercially available Wenger DDC or HIP preconditioners (see U.S. Pat. Nos. 4,752,139, 7,448,795, and 9,028,133). Such preconditioners provide steam and/or water injection in order to heat and pre-cook the starting mixtures, or in some instances hot air may be used as a heating medium (U.S. Pat. No. 7,963,214). In such instances, it may be necessary to provide only a portion of the meat fraction of the mixture during passage through the preconditioner, with separate introduction of the remainder of the meat fraction directly into the input **24** of the processing device, along with the preconditioner materials, to achieve the desired total percentage of meat.

(43) It will be appreciated that in extrusion technology there are two principal sources of energy input, referred to as specific mechanical energy (SME), and specific thermal energy (STE). SME is primarily derived from the heat, friction, and shear forces developed by the extrusion screw(s), whereas STE is generated by the addition of heat exchange media, typically steam. In existing extrusion technology, STE is most commonly added by means of direct injection of steam into the mixture being processed, either in the preconditioner, the extruder, or both. However, as indicated above, direct injection of steam can be problematical when high-meat recipes are being processed. The present invention provides a distinct improvement, in that indirect heating is achieved in the extruder, which avoids direct steam addition and the consequent excess moisture problems attendant thereto.

Claims

1. A method of preparing a food product comprising the steps of: passing a mixture of food ingredients into and through a food preconditioner operable to heat and at least partially pre-cook said food ingredients to create a preconditioned product; passing said preconditioned product into a

food extruder operable to further cook said preconditioned product, said food extruder having a barrel with a preconditioned product inlet, an extruded product output comprising a restricted orifice die, and a pair of elongated, fully intermeshed, axially rotatable screws with outwardly extending helical flighting along the length thereof, the flighting of the screws being fully intermeshed along the lengths thereof, said screws within said barrel and operable to move said mixture from said mixture input toward and through said restricted orifice die, said mixture including grain, starch, and fat, said screws each having a hollow core, and at least a portion of said helical flighting having an internal hollow helical region, said hollow core and region each configured to receive heat exchange media therein, said hollow core and said flighting portion in communication with each other; and during said step of passing the preconditioned product into the food extruder, rotating said screws and generating pressure within said barrel.

2. The method of claim 1, said mixture including meat, said meat present in the mixture at a level of from about 100-250% by weight based upon the weight of the dry ingredients taken as 100% by weight.
3. The method of claim 2, said meat being preheated to a temperature of from about 35-50° C., prior to said step of passing the mixture of food ingredients into and through the food preconditioner.
4. The method of claim 1, during said step of passing the preconditioned product into the food extruder, rotating said screws at a rate of from about 30-150 rpm and developing a pressure within said barrel of from about 200-600 psi, and directing heat exchange media into the hollow cores and hollow helical regions of said screws.
5. The method of claim 1, further comprising directing said heat exchange media into said hollow cores and allowing the heat exchange media to pass into said hollow helical regions.
6. The method of claim 5, said hollow cores and hollow helical regions being in communication through a series of spaced apart apertures, whereby said heat exchange media directed into said hollow cores passes through said apertures and into said hollow helical regions.
7. The method of claim 1, including the step of retarding the flow of said mixture within said barrel at a point proximal to said output.
8. The method of claim 1, said heat exchange media being steam pressurized to a level of from about 30-60 psi.
9. The method of claim 1, including the step of directionally co-rotating said screws within said barrel.
10. The method of claim 1, said food product being a pet food.
11. The method of claim 1, wherein said heat exchange media is delivered to said hollow cores via a media delivery assembly that includes a pair of media delivery tubes extending from a point outside of said barrel and respectively into a corresponding one of said hollow cores.
12. The method of claim 1, wherein each of said screws having a first helical flighting section, the first helical flighting section that conveys said preconditioned product from said barrel inlet toward and through said extruded product outlet, and wherein each of said screws has a second helical flighting section proximal to said extruded product outlet that retards the flow of preconditioned product therepast, said first and second helical sections being of opposite hand.
13. The method of claim 12, said second helical section being shorter than said first helical section.
14. The method of claim 12, said second helical section having a pitch length smaller than the pitch length of said first helical section.
15. The method of claim 12, said first helical section having a pitch length of from about 0.4-1.2 based upon the diameter of said first helical section, and said second helical section having a pitch length of from about 0.2-1.0 based upon the diameter of said second helical section.
16. A method of preparing a food product comprising the steps of: passing a mixture of food ingredients into and through a food preconditioner operable to heat and at least partially pre-cook said food ingredients to create a preconditioned product; passing said preconditioned product into a

food extruder operable to further cook said preconditioned product, said food extruder having a barrel with a preconditioned product inlet, an extruded product output comprising a restricted orifice die, and a pair of elongated, intermeshed, axially rotatable screws with outwardly extending helical flighting along the length thereof, said screws within said barrel and operable to move said mixture from said mixture input toward and through said restricted orifice die, said mixture including grain, starch, and fat, said screws each having a hollow core, and at least a portion of said helical flighting having an internal hollow helical region, said hollow core and region each configured to receive heat exchange media therein, said hollow core and said flighting portion in communication with each other; directing said heat exchange media into said hollow cores and allowing the heat exchange media to pass into said hollow helical regions, said hollow cores and hollow helical regions being in communication through a series of spaced apertures, whereby said heat exchange media directed into said hollow cores passes through said apertures and into said hollow helical regions; and during said passing preconditioned product into a food extruder step, rotating said screws and generating pressure within said barrel.

17. A method of preparing a food product comprising the steps of: passing a mixture of food ingredients into and through a food preconditioner operable to heat and at least partially pre-cook said food ingredients to create a preconditioned product; passing said preconditioned product into a food extruder operable to further cook said preconditioned product, said food extruder having a barrel with a preconditioned product inlet, an extruded product output comprising a restricted orifice die, and a pair of elongated, intermeshed, axially rotatable screws with outwardly extending helical flighting along the length thereof, said screws within said barrel and operable to move said mixture from said mixture input toward and through said restricted orifice die, said mixture including grain, starch, and fat, said screws each having a hollow core, and at least a portion of said helical flighting having an internal hollow helical region, said hollow core and region each configured to receive heat exchange media therein, said hollow core and said flighting portion in communication with each other, wherein said heat exchange media is delivered to said hollow cores via a media delivery assembly that includes a pair of media delivery tubes extending from a point outside of said barrel and respectively into a corresponding one of said hollow cores; and during said passing preconditioned product into a food extruder step, rotating said screws and generating pressure within said barrel.
